

Jenkins Installation Guide for Windows and macOS

Full Step-by-Step Practical Lab Sheet

Prepared for students to follow one line at a time

This guide explains Jenkins installation on Windows and macOS from the beginning to the final dashboard. Each instruction is written stepwise so that beginners can perform the installation without confusion.

Learning Objectives

- Understand the basic purpose of Jenkins.
- Install Java required for Jenkins.
- Install Jenkins on Windows using the official MSI installer.
- Install Jenkins on macOS using Homebrew.
- Unlock Jenkins and complete the first setup wizard.
- Start, stop, and restart Jenkins services.
- Solve common installation problems.

Prerequisites Before Starting

Requirement	Windows	macOS
Operating System	Windows 10 or Windows 11	macOS Monterey or later recommended
RAM	Minimum 4 GB	Minimum 4 GB
Storage	Minimum 10 GB free space	Minimum 10 GB free space
Software Needed	Java JDK and Jenkins MSI	Homebrew, Java JDK, Jenkins LTS
Internet	Required	Required

Introduction to Jenkins

Jenkins is an open-source automation server used in DevOps for Continuous Integration and Continuous Delivery. It helps developers automatically build, test, and deploy software projects.

Jenkins runs on Java, so Java must be installed before Jenkins. After installation, Jenkins opens in a web browser using a local address such as `http://localhost:8080`.

PART A - Jenkins Installation on Windows

A1. Check Windows System Requirements

Step 1: Use Windows 10 or Windows 11.

Step 2: Make sure your computer has at least 4 GB RAM.

Step 3: Make sure your computer has at least 10 GB free storage.

Step 4: Connect your computer to the internet.

Step 5: Login with an administrator account.

A2. Download Java JDK on Windows

Step 1: Open any web browser such as Chrome, Edge, or Firefox.

Step 2: Search for Oracle Java Downloads or open the official Java download page.

Step 3: Choose the Windows x64 installer for JDK.

Step 4: Click Download.

Step 5: Wait until the Java installer file is downloaded.

A3. Install Java JDK on Windows

Step 1: Open the Downloads folder.

Step 2: Double-click the downloaded Java JDK installer.

Step 3: Click Yes if Windows asks for permission.

Step 4: Click Next on the setup screen.

Step 5: Keep the default installation location.

Step 6: Click Next.

Step 7: Wait for the installation to complete.

Step 8: Click Close after installation finishes.

A4. Verify Java Installation on Windows

Step 1: Click the Windows Start button.

Step 2: Search for Command Prompt.

Step 3: Open Command Prompt.

Step 4: Type the command given below.

Step 5: Press Enter.

```
java -version
```

If Java is installed correctly, the Command Prompt shows the installed Java version. Example output may show Java 17 or Java 21.

A5. Download Jenkins on Windows

Step 1: Open a web browser.

Step 2: Go to the official Jenkins download page.

Step 3: Find the Long-Term Support version.

Step 4: Choose Windows.

Step 5: Download the Windows MSI installer.

Step 6: Wait until the Jenkins MSI file is downloaded.

A6. Install Jenkins on Windows

Step 1: Open the Downloads folder.

Step 2: Double-click the Jenkins MSI file.

Step 3: Click Yes if Windows asks for administrator permission.

Step 4: The Jenkins Setup Wizard will open.

Step 5: Click Next.

Step 6: Choose the installation folder or keep the default folder.

Step 7: Click Next.

Step 8: Choose Run service as LocalSystem for a simple student setup.

Step 9: Click Next.

Step 10: Keep the default Jenkins port as 8080.

Step 11: Click Test Port if the option is shown.

Step 12: If the port is available, click Next.

Step 13: The installer will detect Java automatically.

Step 14: Click Next.

Step 15: Click Install.

Step 16: Wait until the installation completes.

Step 17: Click Finish.

A7. Open Jenkins on Windows

Step 1: Open a web browser.

Step 2: Click the address bar.

Step 3: Type the Jenkins local URL shown below.

Step 4: Press Enter.

```
http://localhost:8080
```

The Jenkins Unlock Jenkins page should open. If the page does not open, check whether the Jenkins service is running.

A8. Find Initial Admin Password on Windows

Step 1: Open File Explorer.

Step 2: Go to the C drive.

Step 3: Open the ProgramData folder.

Step 4: If ProgramData is hidden, enable Hidden items from the View menu.

Step 5: Open the Jenkins folder.

Step 6: Open the .jenkins folder.

Step 7: Open the secrets folder.

Step 8: Open the file named initialAdminPassword using Notepad.

Step 9: Copy the password from the file.

```
C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword
```

A9. Unlock Jenkins on Windows

Step 1: Return to the Jenkins browser page.

Step 2: Paste the copied password into the Administrator password box.

Step 3: Click Continue.

Step 4: Wait for Jenkins to move to the plugin installation screen.

A10. Install Suggested Plugins on Windows

Step 1: On the Customize Jenkins screen, click Install suggested plugins.

Step 2: Wait while Jenkins downloads and installs plugins.

Step 3: Do not close the browser during plugin installation.

Step 4: Wait until the plugin installation process is complete.

Suggested plugins usually include important plugins such as Git, Pipeline, Credentials, SSH, and other commonly used Jenkins tools.

A11. Create First Admin User on Windows

Step 1: Enter a username.

Step 2: Enter a strong password.

Step 3: Confirm the password.

Step 4: Enter full name.

Step 5: Enter email address.

Step 6: Click Save and Continue.

Step 7: Keep the Jenkins URL as `http://localhost:8080` unless your instructor says otherwise.

Step 8: Click Save and Finish.

Step 9: Click Start using Jenkins.

A12. Confirm Jenkins Dashboard on Windows

Step 1: Check that the Jenkins Dashboard is visible.

Step 2: Find the New Item option on the left side.

Step 3: Find the Manage Jenkins option on the left side.

Step 4: Find Build Queue and Build Executor Status on the dashboard.

Step 5: This confirms Jenkins installation is complete on Windows.

A13. Start, Stop, and Restart Jenkins Service on Windows

Step 1: Press Windows + R.

Step 2: Type `services.msc`.

Step 3: Press Enter.

Step 4: Find Jenkins in the services list.

Step 5: Right-click Jenkins.

Step 6: Click Start to start Jenkins.

Step 7: Click Stop to stop Jenkins.

Step 8: Click Restart to restart Jenkins.

```
services.msc
```

A14. Windows Troubleshooting

Problem 1 - Jenkins page does not open

Step 1: Open Services using `services.msc`.

Step 2: Find Jenkins.

Step 3: Check whether the service status is Running.

Step 4: If it is not running, right-click Jenkins and click Start.

Step 5: Open `http://localhost:8080` again.

Problem 2 - Port 8080 is already used

Step 1: Open Command Prompt as administrator.

Step 2: Run the command below.

Step 3: Find which process is using port 8080.

Step 4: Stop that process or change the Jenkins port.

```
netstat -ano | findstr :8080
```

Problem 3 - Java not detected

Step 1: Open Command Prompt.

Step 2: Run `java -version`.

Step 3: If Java is not recognized, reinstall Java JDK.

Step 4: Restart the computer.

Step 5: Run Jenkins installer again if required.

PART B - Jenkins Installation on macOS

B1. Check macOS System Requirements

Step 1: Use macOS Monterey or later if possible.

Step 2: Make sure your Mac has at least 4 GB RAM.

Step 3: Make sure your Mac has at least 10 GB free storage.

Step 4: Connect your Mac to the internet.

Step 5: Make sure you know your Mac login password.

B2. Open Terminal on macOS

Step 1: Press Command + Space.

Step 2: Type Terminal.

Step 3: Press Enter.

Step 4: The Terminal window will open.

B3. Install Homebrew on macOS

Homebrew is a package manager used to install software from the Terminal on macOS.

Step 1: Copy the command below.

Step 2: Paste it into Terminal.

Step 3: Press Enter.

Step 4: Enter your Mac password if asked.

Step 5: Wait until Homebrew installation completes.

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

B4. Verify Homebrew Installation

Step 1: Open Terminal.

Step 2: Type the command below.

Step 3: Press Enter.

```
brew --version
```

If Homebrew is installed correctly, Terminal will show the Homebrew version.

B5. Install Java on macOS

Step 1: Open Terminal.

Step 2: Type the command below.

Step 3: Press Enter.

Step 4: Wait until Java installation completes.

```
brew install openjdk
```

B6. Verify Java on macOS

Step 1: Open Terminal.

Step 2: Type the command below.

Step 3: Press Enter.

```
java -version
```

If Java is installed correctly, Terminal displays the installed Java version.

B7. Install Jenkins LTS on macOS

Step 1: Open Terminal.

Step 2: Type the command below.

Step 3: Press Enter.

Step 4: Wait for Homebrew to download and install Jenkins LTS.

```
brew install jenkins-lts
```

B8. Start Jenkins on macOS

Step 1: Open Terminal.

Step 2: Type the command below.

Step 3: Press Enter.

Step 4: Wait a few seconds for Jenkins service to start.

```
brew services start jenkins-lts
```

B9. Check Jenkins Service Status on macOS

Step 1: Open Terminal.

Step 2: Run the command below.

Step 3: Check that jenkins-lts shows started.

```
brew services list
```

B10. Open Jenkins on macOS

Step 1: Open Safari, Chrome, or Firefox.

Step 2: Click the address bar.

Step 3: Type the Jenkins local URL below.

Step 4: Press Enter.

```
http://localhost:8080
```

The Jenkins Unlock Jenkins page should open in the browser.

B11. Find Initial Admin Password on macOS

Step 1: Open Terminal.

Step 2: Type the command below.

Step 3: Press Enter.

Step 4: Copy the password shown in Terminal.

```
cat ~/.jenkins/secrets/initialAdminPassword
```

B12. Unlock Jenkins on macOS

Step 1: Return to the Jenkins browser page.

Step 2: Paste the copied password into the Administrator password box.

Step 3: Click Continue.

Step 4: Wait for Jenkins to open the plugin installation screen.

B13. Install Suggested Plugins on macOS

Step 1: Click Install suggested plugins.

Step 2: Wait until Jenkins downloads plugins.

Step 3: Do not close the browser.

Step 4: Continue after all plugins are installed.

B14. Create Admin User on macOS

Step 1: Enter username.

Step 2: Enter password.

Step 3: Confirm password.

Step 4: Enter full name.

Step 5: Enter email address.

Step 6: Click Save and Continue.

Step 7: Keep Jenkins URL as http://localhost:8080.

Step 8: Click Save and Finish.

Step 9: Click Start using Jenkins.

B15. Confirm Jenkins Dashboard on macOS

Step 1: Check that Jenkins Dashboard is visible.

Step 2: Find New Item on the left side.

Step 3: Find Manage Jenkins on the left side.

Step 4: Confirm Jenkins installation is complete on macOS.

B16. Stop Jenkins on macOS

Step 1: Open Terminal.

Step 2: Run the command below.

Step 3: Wait until Jenkins service stops.

```
brew services stop jenkins-lts
```

B17. Restart Jenkins on macOS

Step 1: Open Terminal.

Step 2: Run the command below.

Step 3: Wait until Jenkins restarts.

Step 4: Open <http://localhost:8080> again.

```
brew services restart jenkins-lts
```

B18. macOS Troubleshooting

Problem 1 - Jenkins does not open

Step 1: Open Terminal.

Step 2: Run brew services list.

Step 3: Check whether jenkins-lts is started.

Step 4: If Jenkins is stopped, run `brew services start jenkins-lts`.

Step 5: Open <http://localhost:8080> again.

Problem 2 - Port 8080 is already used

Step 1: Open Terminal.

Step 2: Run the command below.

Step 3: Find the process using port 8080.

Step 4: Stop the process or change Jenkins port.

```
lsof -i :8080
```

Problem 3 - Homebrew command not found

Step 1: Restart Terminal.

Step 2: Check whether Homebrew installation completed successfully.

Step 3: Run `brew --version`.

Step 4: If command is still not found, reinstall Homebrew carefully.

PART C - Jenkins Folder Locations

Item	Windows Location	macOS Location
Jenkins Home	C:\ProgramData\Jenkins\jenkins	~/.jenkins
Jobs Folder	C:\ProgramData\Jenkins\jenkins\jobs	~/.jenkins/jobs
Plugins Folder	C:\ProgramData\Jenkins\jenkins\plugins	~/.jenkins/plugins
Secrets Folder	C:\ProgramData\Jenkins\jenkins\secrets	~/.jenkins/secrets

Workspace Folder	C: \ProgramData\Jenkins\jenkins\workspace	~/.jenkins/workspace
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PART D - Important Jenkins UI Options After Installation

D1. Dashboard

Step 1: Open <http://localhost:8080>.

Step 2: Login using your admin username and password.

Step 3: The dashboard shows all Jenkins jobs.

Step 4: The dashboard also shows build queue and executor status.

D2. New Item

Step 1: Click New Item.

Step 2: Enter a job name.

Step 3: Choose Freestyle project or Pipeline.

Step 4: Click OK to create the job.

D3. Manage Jenkins

Step 1: Click Manage Jenkins.

Step 2: Use this section for system settings.

Step 3: Use this section for plugins.

Step 4: Use this section for security.

Step 5: Use this section for credentials and nodes.

D4. Plugins

Step 1: Click Manage Jenkins.

Step 2: Click Plugins.

Step 3: Use Available Plugins to install new plugins.

Step 4: Use Installed Plugins to check existing plugins.

Step 5: Use Updates to update installed plugins.

D5. Credentials

Step 1: Click Manage Jenkins.

Step 2: Click Credentials.

Step 3: Use credentials to store usernames, passwords, SSH keys, and tokens securely.

PART E - Recommended Plugins for Students

Plugin	Purpose
Git Plugin	Connect Jenkins with Git repositories
Pipeline Plugin	Create Jenkins pipeline jobs
Credentials Plugin	Store secrets safely

SSH Plugin	Connect remote servers
Docker Plugin	Use Docker with Jenkins
Role Strategy Plugin	Manage role-based access

PART F - Basic Security After Installation

- Step 1:** Do not share the Jenkins admin password.
- Step 2:** Create separate users for each student or team member.
- Step 3:** Disable anonymous access.
- Step 4:** Install only trusted plugins.
- Step 5:** Update Jenkins and plugins regularly.
- Step 6:** Use strong passwords.
- Step 7:** Use credentials instead of writing passwords directly in jobs.
- Step 8:** Take backup of the Jenkins home folder.

PART G - Final Verification Checklist

- Step 1:** Java is installed successfully.
- Step 2:** `java -version` command works.
- Step 3:** Jenkins is installed successfully.
- Step 4:** Jenkins opens at `http://localhost:8080`.
- Step 5:** Initial admin password is copied correctly.
- Step 6:** Jenkins is unlocked.
- Step 7:** Suggested plugins are installed.
- Step 8:** Admin user is created.
- Step 9:** Jenkins dashboard is visible.
- Step 10:** Jenkins service can be started and stopped.

Conclusion

Jenkins installation on Windows and macOS becomes simple when each step is followed in sequence. Windows users mainly install Jenkins using the MSI installer, while macOS users commonly install Jenkins using Homebrew. After completing installation, students should understand how to open Jenkins, unlock Jenkins, install plugins, create the first admin user, and manage the Jenkins service.